

Towards the European Business Wallet

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Abstract: Against the background of the recent announcement of the European Commission to create a “European Business Wallet” (EUBW), which aims at facilitating trustworthy electronic business and government processes, the present paper recalls the relevant regulatory and technical background for the EUBW, contributes to the necessary standardisation process for legal person identities and outlines how the EUBW might in the end look like from a technical perspective.

Keywords: EUDI, EUDIW, EUBW, eIDAS, Wallet, Verifiable Credential (VC)

1 Introduction

On January 29th 2025 Ursula von der Leyen presented the European Commission’s plan to boost Europe’s competitiveness and turn it into a place where future technologies, services, and clean products are invented, manufactured, and put on the market. The presented “Competitiveness Compass” [EC25a; EC25b] serves as roadmap and sets out the course the Commission will follow in the next five years. Among the announced “Flagship Actions enablers” for the year 2025 is the “European Business Wallet” (EUBW), which will build upon the amended eIDAS-Regulation (EU) No. 910/2014 and is envisioned to be “the cornerstone of doing business simply and digitally in the EU, providing a seamless environment for companies to interact with all public administrations.”

While the closely related “European Digital Identity Wallet” (EUDIW) is defined in Art. 3 (42) of (EU) No. 910/2014 to be “an electronic identification means which allows the user to securely store, manage and validate person identification data and electronic attestations of attributes for the purpose of providing them to relying parties and other users of European Digital Identity Wallets, and to sign by means of qualified electronic signatures or to seal by means of qualified electronic seals”, it is not entirely clear yet, what the announced EUBW exactly will be from a technical perspective.

But it is already very clear today, that EUDIW-like solutions for server-based environments and related trust services for the management of machine-readable legal person identities will play a crucial role for the envisioned trustworthy automation of electronic business and government processes across Europe and hence the present paper aims at carrying together

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aspects, which seem to be relevant for paving the way for the upcoming design phase of the EUBW and related standardisation activities.

Section 4 briefly outlines how the EUBW may look from a technical perspective and how it may be integrated into the existing and emerging system landscape and Section 5 finally summarises the present paper and provides a brief outlook towards possible future developments.

Note: This document is an *extended abstract* only to meet the given page limits. Please reach out to the lead author to request the full version of the paper and do not hesitate to contribute to the “EU Business Wallet Initiative” initiated by the non-profit [go.eIDAS Association](#).

2 Background for European Business Wallets

2.1 Relevant Regulatory Framework

2.1.1 eIDAS-Regulation (EU) No. 910/2014

Among the most relevant legal acts for the EUBW is certainly the so called “eIDAS-Regulation” (EU) No. 910/2014, which introduces the closely related EUDIW in its Art. 5a. While the eIDAS-Regulation formally addresses both natural and legal persons and defines the relevant terms in Art. 3 in a way, which covers both natural and legal persons, looking at the details of the first batch of endorsed implementing acts⁴ and the currently available “Architecture and Reference Framework” [ARF], which is part of the common “Union Toolbox” recommended in (EU) 2021/946 for a coordinated approach towards the “European Digital Identity Framework”, it becomes clear that the EUDIW is – at least until now – mainly geared to be used in processes for natural persons using mobile phones.

This claim is supported by the fact that the current “Wallet Reference Implementation” is only available for Android and iOS as well as by the simple observation, that the technical details for the “Person Identification Data” (PID) for legal persons, called “Legal-person PID” or simply “LPID” in the following, are not yet defined in a precise technical manner, which would give rise for interoperable implementations.

While the *legal* definition of PID in Art. 3 (3) of (EU) No. 910/2014 is “a set of data that is issued in accordance with Union or national law and that enables the establishment of the identity of a natural or legal person, or of a natural person representing another natural person or a legal person”, and therefore covers *both* natural and legal persons, the *technical* specification in the Annex of (EU) 2024/2977 only contains “data identifiers”, or “claim names”, for natural persons yet. As the Tables 3 and 4 of the Annex of (EU) 2024/2977 for

⁴ See (EU) 2024/2977, (EU) 2024/2979, (EU) 2024/2980, (EU) 2024/2981 and (EU) 2024/2982.

LPID unfortunately do not yet contain the necessary “data identifiers” for legal persons, it is currently not possible to implement systems issuing, handling and presenting LPID in an interoperable manner.

In a similar manner, the so called “PID Rulebook” in [Annex 3.1](#) in the [\[ARF\]](#) does – at least currently – not address legal persons at all and “Topic 28 (Wallet Unit for legal persons)” in [Annex A.2.3.28](#) of the [\[ARF\]](#) only states that

- “Legal-person PIDs” and “Wallet Units for legal persons” will be discussed with the Member States in the future,
- a “Legal-person PID” is issued under an eID scheme⁵ and that
- the Commission shall develop a “Legal-person PID Rulebook” to specify the attribute scheme and other technical details applicable for “Legal-person PIDs.”

The current list of notified eID schemes is available in the Official Journal of the European Union [\[OJ24\]](#) and the related notification documents are available in the [Wiki of the eIDAS Collaboration Network](#). As only the eID schemes in Austria, Bulgaria, Croatia and the Netherlands seem to involve legal person identities today, it will be interesting to see whether the EUBW-related legislation will introduce changes here in order to facilitate the issuance and use of LPID across Europe.

We will come back to this topic in [Section 3](#), when we outline, how an LPID may look like from a technical point of view.

2.1.2 Regulations related to legal entity identifiers

As the LPID will contain one or more legal entity identifiers, it is worthwhile to consider the existing regulations, which define or use such legal entity identifiers, as listed in [Table 4](#) within the Annex of [\(EU\) 2024/2977](#) and in a similar manner also in the [draft of the implementing act according to Article 5b \(11\) of \(EU\) No. 910/2014](#):

- Value Added Tax Identification Number (VATIN) according to the [Council Directive 2006/112/EC](#)
- [Taxpayer Identification Number \(TIN\)](#)
- European Unique Identifier (EUID) according to [\(EU\) 2017/1132](#), [\(EU\) 2020/2244](#) and [\(EU\) 2021/1042](#), which is based on ISO/IEC 6523-1 [\[ISO23\]](#)

⁵ Note, that this statement is perfectly in line with the eIDAS-Regulation, as [Art. 5a \(11\) of \(EU\) No 910/2014](#) states that “European Digital Identity Wallets shall be provided under an electronic identification scheme with assurance level high.” This stipulation is somewhat unfortunate for the EUBW because there is no differentiation with respect to identities for natural and legal persons here and most use cases for EUBW only require a substantial level of security, which can be achieved without special purpose cryptographic hardware.

- Legal Entity Identifier (LEI) according to (EU) No 2022/1860, which is based on ISO 17442-1 [ISO20a]
- Economic Operator Registration and Identification Number (EORI-No) according to (EU) No 1352/2013
- Excise Number provided in Article 2(12) of Council Regulation (EU) No 389/2012

Furthermore the recently enacted directive⁶ (EU) 2025/25, which amends Directive (EU) 2017/1132, should be taken into account in Section 3 and for the standardisation of a future legal entity identity to be used together with the EUBW.

The novel Directive (EU) 2025/25 outlines in Art. 16b the “EU Company Certificate” for limited liability companies (Nr. 2) and partnerships (Nr. 3). According to (Nr. 6), the issued “EU Company Certificate” shall be authenticated by means of trust services according to (EU) No 910/2014 in a way, which is compatible with the EUDIW.

In a similar manner Art. 16c of this Directive outlines the “Digital EU power of attorney”, which authorises a person to represent the company and which shall also be authenticated by means of trust services according to (EU) No 910/2014 while the granting, amending or revocation of the power of attorney shall be compatible with the EUDIW.

2.1.3 Other regulations with relevance for the EU Business Wallet

In addition to the eIDAS-Regulation (see Section 2.1.1) and the regulations related to legal entity identifiers (see Section 2.1.2) there are numerous additional regulations which are relevant for the EUBW, because they introduce *application specific requirements for identification and authentication* of business partners, including consumers as well as other enterprises or government authorities, or data objects with varying security and policy requirements.

We only provide a few first examples here and refer to the full version of the paper for a more systematic treatment of the existing EU regulations in the 32 policy areas of [EC25c]:

- *Second Payment Services Directive (PSD2)* (EU) 2015/2366 requires in Art. 34 of its implementing act (EU) 2018/389 to use qualified certificates for website authentication and/or qualified certificates for electronic seals for the authentication of payment service providers.
- *Anti Money Laundering Regulation (AMLR)* (EU) 2024/1624 requires in its Art. 22 Nr. 6. that obliged entities shall identify their customers, or any person acting on their behalf, by either using a conventional identity document or passport or by electronic

⁶ According to Art. 4 of (EU) 2025/25 the EU Member States shall adopt and publish, by 31 July 2027, the laws, regulations and administrative provisions necessary to comply with this Directive and apply the measures from 31 July 2028.

identification means according to the eIDAS-Regulation (EU) No 910/2014 with assurance levels ‘substantial’ or ‘high’ or relevant qualified trust services according to this regulation to prevent the use of the financial system for the purposes of money laundering or terrorist financing.

- *Energy Labelling Regulation* (EU) 2017/1369 requires in Art. 4 of its implementing act (EU) 2024/994 to use qualified electronic seals for adding data to the “European Product Registry for Energy Labelling” (EPREL) data base.
- *Ecodesign for Sustainable Products Regulation* (EU) 2024/1781 requires in Art. 11 that the “Digital Product Passport” ensures authentication, reliability and integrity with a high level of security and privacy.
- *Single Digital Gateway (SDG) Regulation* (EU) 2018/1724 requires in Art. 13 that cross-border users are able to identify and authenticate itself and sign or seal documents electronically based on means of the eIDAS-Regulation.
- *European Health Data Space (EHDS)* (EU) 2025/327 requires that recognised eID-means according to Art. 6 of (EU) No. 910/2014 shall be used by Health Professionals (Art. 12) and Citizen (Art. 16) for health data access.
- *Data Governance Act* (EU) 2022/868 introduces in Art. 2 (20) a so called “Secure Processing Environment”, which requires confidentiality, integrity and accessibility of data.
- *Electronic Freight Transport Information (eFTI)* (EU) 2020/1056 requires in Art. 4 that information in a machine-readable format shall be made available via an authenticated and secure connection.
- *électronique Convention relative au contrat de transport international de Marchandises par Route (eCMR)* is the related electronic consignment note specified by the “Open Logistics Foundation”, which uses advanced electronic seals according to Art. 36 of (EU) No. 910/2014 to enable trustworthy electronic bills of lading.

2.2 Most Relevant Standards and Specifications

This section lists some of the seemingly most relevant standards and specifications for the EU Business Wallet. While a complete survey is beyond the scope of the present paper, it is nevertheless hoped that this section might provide a first starting point for a more systematic survey as basis for the upcoming design and standardisation phase.

2.3 Standards for Machine Readable Data Formats

As the EUBW is envisioned to automate business and government processes and exchange data in a machine readable fashion, it is obvious that standards like JSON [Br17], XML [XML], CBOR [BH20] and ASN.1 [ASN.1] are relevant for the EUBW.

2.4 Standards for Application Programming Interfaces and Business Data Exchange

Based on the machine readable data formats above, there are standards to create “Application Programming Interfaces” (API) and web services, which include *OpenAPI*, *AsyncAPI*, *SOAP*, *ebXML* and protocols like the *Dataspace Protocol*.

2.5 Standards for Credential Management and Authorisation

For the identification, authentication, authorisation and related security services, the participants in a communication need appropriate credentials. The most important credential standards available today seem to be as follows:

- [X.509] certificates represent the most classical form of credential, which is issued by a “Certification Authority” in order to bind the public key of an entity to its identity.
- Within the XML-based era [SAML]-based credentials played a major role, whereas an “Identity Provider” issued so called “Assertions” about a subject, which typically includes a set of attributes. For cross-border eID-use there is the [eID] profile, which seems to be worth to consider for the specification of identities for legal entities and for the specification of LPID in Section 3.
- Today [SAML] is about to be replaced by the more lightweight protocol stack based on OAuth [Ha12] and “OpenID Connect” (OIDC) [SB+23], in which an “OpenID Provider” issues JSON-based [Br17] “ID Tokens”, which are in fact “JSON Web Tokens” (JWT) [JWT].
- The classical “Federated Identity Management” architecture is about to be replaced by architectures based on the “Self-Sovereign Identity” (SSI) paradigm. Here an “Issuer” issues “Verifiable Credentials” [W3C22; W3C25b] based on JSON [Br17], JSON-LD [SL+20] or CBOR [BH20] within the “mdoc” format for the mobile drivers licenses (mDL) according to ISO/IEC 18013-5 [ISO21], for example. There is also the option that the credentials support “Selective Disclosure” (SD-JWT [FYC25] and SD-JWT VC [TFC24]). The most important protocol for the issuance of credentials to the wallet today is “OpenID for Verifiable Credential Issuance” (OID4VCI) [LY+24] and for the presentation of claims to the “Verifier” there are protocols like “OpenID for Verifiable Presentations” (OID4VP) [TL25] and DIDComm [CLT].
Currently there are at least the following different credentials formats:
 - JSON-based and signed as JWT [JWT] according to the W3C Verifiable Credential Data Model (VCDM) [W3C22; W3C25b],
 - JSON-based and prepared for supporting selective disclosure and signed as JWT [JWT] (SD-JWT [FYC25] and SD-JWT VC [TFC24]),

- CBOR-based and protected with corresponding “CBOR Object Signing and Encryption” (COSE) [Sh22] mechanisms, as used in ISO/IEC 18013-5 [ISO21] for example,
- JSON-LD-based and signed as JWT [JWT] according to the W3C VCDM [W3C22; W3C25b] or
- protected by some alternative “Data Integrity Proof” [W3C25c] using “Controlled Identifiers” (CIDs) [LS+25] or “Decentralized Identifiers” (DIDs) [SL+22] with a suitable “DID-Method” [W3C25a]. This approach is used in some European communities with “DID:EBSI” with first “Business Wallets” and related use cases within the “European Blockchain Services Infrastructure” [EB25a; EB25b; EB25c].

An alternative approach, which is not yet fully standardised as basic mechanism, but nevertheless has been used for the creation of the “Verifiable Legal Entity Identifier” (vLEI) according to [ISO24] is to use the “Key Event Receipt Infrastructure” [KERI] and “Authentic Chained Data Containers” [SF] for the creation of credentials.

3 Towards Standards for Legal Person Identities

In order to create the missing standard(s) for Legal Person Identities, we propose the following approach, which consists of three simple steps, which are explained in more detail in the following sub-sections:

1. Create abstract data model for legal person identities
2. Specify the mapping of the data model to standardised credential formats
3. Define the details of the trust framework for the management of the legal person identities

3.1 Data Model for Legal Person Identities

In a first step we create an abstract data model for legal entities and legal persons based on the existing regulations related to identifiers as outlined in Section 2.1.2 considering previous work [SEM24; Sp24] and in particular the recent work of the European Commission [EC-TS2] for the notification of wallet-related providers according to the amended eIDAS-regulation (EU) No. 910/2014 and the specific implementing act (EU) 2024/2980.

In addition to the core elements already present in [EC-TS2], a LegalPerson may have additional attributes as part of its envisioned “EU Company Certificate” according to (EU) 2025/25, as explained in the full paper.

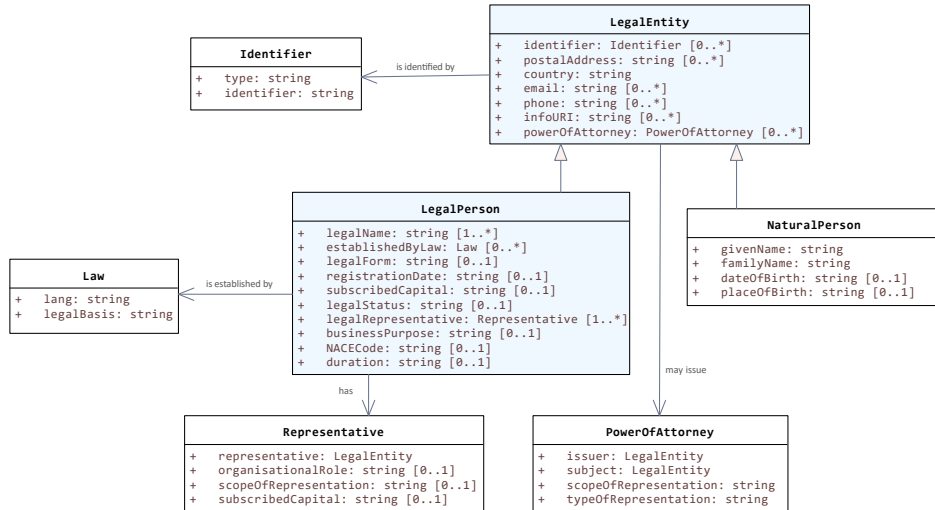


Fig. 1: Data Model for Legal Person Identities

3.2 Mapping of Data Model to Standardised Credential Formats

The abstract data model outlined above may, of course, be serialised to any machine readable data format mentioned in Section 2.3. Furthermore, it may also be mapped to any of the standardised credential formats listed in Section 2.5 in a more or less straightforward manner.

Only the mapping to [X.509] and related standards produced by ISO, ITU, IETF and last but not least the standards produced by ETSI ESI seems to be worth to be explicitly explained here in more detail, as these certificates are already widely used within the electronic identification schemes and the trust services according to the eIDAS-Regulation (EU) No. 910/2014 and also in the envisioned trust framework around the EUDIW according to (EU) 2024/2980.

Note, that the issuance of an LPID in form of a suitable [X.509] certificate, which can be used for the creation of at least advanced electronic seals, allows that the *subject* of the certificate can, more or less immediately, act as *issuer* of electronic attestations of attributes according to Art. 3 (EU) No. 910/2014, provided that such an “Enterprise EUBW” supports the functionality for issuing credentials using suitable protocols, such as OID4VCI [LY+24] for example.

Note, that [ISO20b] defines a specific *roleExtension*, which can be used for the encoding of the *organisationalRole* attribute, but some other attributes from the data model depicted in Figure 1 do not seem to be specified in international standards yet.

LegalPerson attribute	[X.509] field	Reference(s)
legalName	organizationName commonName	[ET25d, § 4.2.1] [ET25d, § 4.2.1]
establishedByLaw	QcPSB (esi4-qcStatement10)	[ET25e] with CR5
identifier	organizationIdentifier	[ET25d, § 4.2.1] and [ET25c, § 5.1.4] with CR7, see also Section 2.1.2
postalAddress	postalAddress	[X.520, § 6.6.1]
country	countryName	[ET25d, § 4.2.1]
email	rfc822Name	[X.509, § 9.3.2.1]
phone	telephoneNumber	[X.520, § 6.7.1]
infoURI	uniformResourceIdentifier	[X.509, § 9.3.2.1]

Tab. 1: Mapping of LegalPerson attributes to [X.509]

3.3 Trust Framework for Legal Person Identities

The “canonical trust framework“ for Legal Person Identities used by the EUBW is defined by the eIDAS-Regulation (EU) No. 910/2014, which both regulates *electronic identification (eID)* (see Section 3.3.1) for natural and legal persons as well as *trust services* (Art. 3 (16)) (see Section 3.3.2).

Furthermore, we will briefly outline the main currently available options for EU Member States for the issuance of LPID in Section 3.3.3.

3.3.1 Electronic Identification

The legal framework for eID is more or less unchanged since 2014 and contains in particular the concept of *notification* (Art. 9), which involves a *peer-review* (Art. 12 (5)) of the electronic identification scheme (Art. 3 (4)) by the other EU Member States and finally leads to the *mutual recognition* (Art. 9) of the involved electronic identification means (Art. 3 (2)).

In this domain the EU Member States are more or less free to use their favorite technologies, the trust is induced by the scrutiny of the rather sophisticated and laborious peer-review process and the interoperability is induced, at least in theory, by the [SAML]-based interoperability framework (Art. 12 and the implementing act (EU) 2015/1501).

3.3.2 Trust Services

The initial eIDAS-Regulation from 2014 has replaced the infamous “Electronic Signature Directive” 1999/93/EG and in particular already contained the concept of qualified and

non-qualified trust services for the issuance of [X.509] certificates to natural persons for advanced electronic signatures (Art. 26) and to legal persons for advanced electronic seals (Art. 36).

The amendment in 2024 has introduced the EUDIW (Art. 5a) and additional trust services, which are relevant for the EUBW, such as the services for the electronic attestations of attributes (Art. 45b-45h), for example.

Here, one may distinguish between

- simple *electronic attestations of attributes* (EEA) (Art. 3 (44)),
- well-regulated *qualified electronic attestations of attributes* (QEEA) (Art. 3 (45)) and
- *electronic attestations of attributes issued by or on behalf of a public sector body responsible for an authentic source* (PubEEA) (Art. 3 (46)), which aim at a similar level of trust as QEEA, but directly involve the authentic sources (Art. 3 (47)), which are usually provided by public sector bodies.

The trust services for electronic attestation of attributes may be combined with other trust services including the newly introduced trust service for electronic ledger (Art. 45i), as provided by the “European Blockchain Services Infrastructure” [EB25a], for example. This approach has been followed in projects such as EBSI-VECTOR and EBSI-Trace4EU for the issuance and usage of LPID and the management of Powers of Attorney by responsible natural persons.

3.3.3 Options for the issuance of LPID and their use in the EUBW

The available options for the issuance of LPID may be summarised as follows:

- EU Member States, which have not done this yet, may become active in order to integrate aspects related to legal person identities into their existing eID schemes with a high level of assurance (see Art. 5a (11)) and amend the corresponding notifications (see [OJ24]). As soon as this has happened, they can designate one or more LPID issuers, which may integrate suitable authentic sources, as foreseen in Art. 3 of (EU) 2024/2977, Annex II Nr. 3 of (EU) 2024/2980 and [EC25d].
- On the other hand, an EU Member State may leave the issuance of LPID more or less to the market of (qualified) trust service providers, which may in turn may collaborate with business registries, so that the management of LPID would essentially become a (qualified) trust service for (Q)EAA.
- A third and somewhat intermediate strategy would use PubEEA-providers for the issuance of LPID based on direct access to the relevant registers and authentic sources (Art. 3 (47)).

It is not surprising that the different options for establishing a dedicated LPID issuer, which may be a (Q)TSP or PubEAA issuer, are currently subject to a variety of large scale pilot projects across Europe. It will be interesting to see, which strategies are in the end preferred by the different EU Member States, what the timeline for the individual actions will be, what influence any changes of the legal framework due to the announced EUBW may have and last but not least, which impact the forthcoming standard ETSI TS 119 479 [ET25b] for the “*Specification of interfaces related to Authentic Sources*” may have on these exciting developments.

4 Solution Outline for a European Business Wallet

This section sketches how the EUBW might look like in the end from a technical perspective.

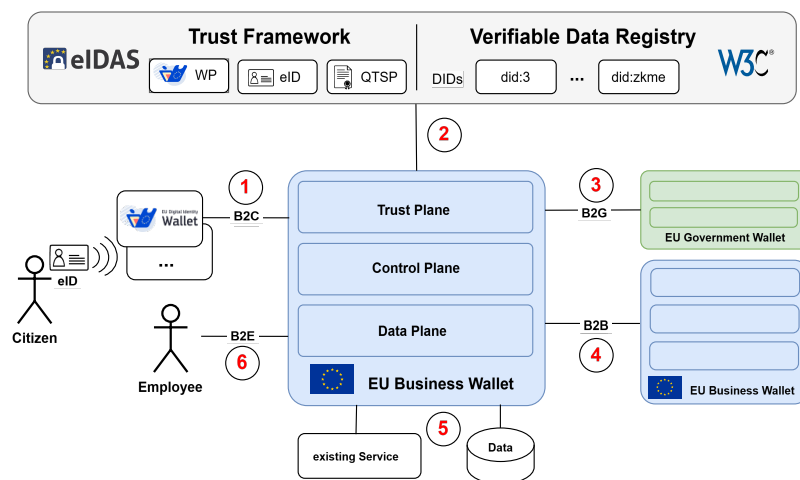


Fig. 2: Outline of System Architecture with EU Business Wallet

As outlined in the Figure 2, the EUBW will (1) support B2C processes together with the EUDIW and hence the EUBW shall be able to act as “Wallet-Relying Party” and use “Access Certificates” in protocols like OID4VP [TL25] and DIDComm [CLT] for example.

The EUBW will (2) request the necessary “Access Certificates” as well as further potentially required certificates and attestations (e.g. LPID) and interact with other suitable trust service providers according to the eIDAS-Regulation (EU) No. 910/2014.

The EUBW will (3) hopefully also be usable for electronic government processes, as soon as there are “EU Government Wallets” around, which support normalised interfaces and standards according to the state of the art.

An EUBW-instance will, of course, be able to conduct trustworthy electronic business processes with other EUBW-instances (4) and will need to smoothly integrated with other business IT systems, as well as existing services (5) and finally support B2E processes, which involve employees (6).

5 Conclusion and Outlook

The present document aims at contributing to the standardisation of the recently announced EUBW by providing a compact survey of the probably most relevant European regulations (see Section 2.1) and seemingly relevant international technical standards (see Section 2.2), before we address Legal Person Identities in Section 3, which are crucial for the successful use of the EUBW. The paper concludes in Section 4 by outlining a technical solution for the envisioned EUBW.

The EUBW is geared towards supporting the most important European initiatives and regulations mentioned above, as well as a large variety of hardly regulated business processes, such as the “Digital Product Passport” or “Data Spaces”, which are emerging for the secure and trustworthy sharing of sensitive data in different industries. For this purpose it is of utmost importance to define a modular and coherent European standardisation framework to achieve these goals and contribute to the design, development and operation of a trustworthy and sovereign “EuroStack”.

The present paper has already been provided as input to the “EU Business Wallet Initiative”, which aims at further paving the way for the EUBW by incubating standardisation activities at the European and later international level.

While the exciting development around the EUBW is in a rather early phase, we already thankfully acknowledge, that the present paper has greatly benefited from very fruitful discussions with numerous experts in pertinent national and international projects and standardisation bodies, which are currently working on the standardisation of a trustworthy EUDIW and LPID.

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